

No Finite Separable Representation at Three Prime Generators

Nonsingular minors of every order, two-prime Hecke–Mahler polynomials, and the open boundary for Erdős Problem #269

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ABSTRACT

For three pairwise distinct primes, the reciprocal running-LCM kernel has nonsingular minors of every order, with one choice of indices valid in every third-coordinate layer. Dividing out row and column powers leaves a binary carry whose consecutive threshold columns differ on disjoint intervals. This gives both arbitrary-order non-separability and the exact rank of every finite sample. The two-prime sums are polynomials in one transcendental Hecke–Mahler value. The repeated three-prime irrationality question remains open.

Let P be a finite set of at least two primes and $\mathcal{S}_P$ its positive smooth integers. Put

$$L(x) = \text{lcm}\{u \in \mathcal{S}_P : u \leq x\}, \quad \mathcal{R}_P = \sum_{u \in \mathcal{S}_P} L(u)^{-1}.$$

Erdős asks whether $\mathcal{R}_P$ is irrational [1, p. 65][2, p. 106]. For $P = \{p, q, r\}$, define

$$H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor}, \quad K(i, j, k) = H(p^i q^j r^k)^{-1}.$$

Prime-power divisibility gives $L = H$ (Proposition 3.1).

1 The binary carry and arbitrary-order rank

Theorem 1.1 (arbitrary-order non-separability). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j, k)$, $0 \leq \ell < d$, satisfying

$$K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k) \quad \text{for all } i, j, k.$$

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Proof. Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$. Splitting the floor exponents gives

$$K(i, j, k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \quad C_{ij} = c^{\mathbf{1}_{\{x_i + y_j \geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q(p^i r^k) \rfloor} r^{k + \lfloor \log_r p^i \rfloor},$$

$$V_j(k) = q^j p^{\lfloor \log_p(q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

Both factors are positive. Distinct primes make $\log_r p$ and $\log_r q$ irrational, so each individual rotation is dense. For $n \geq 1$, choose distinct row indices with $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. Choose $1 - y_{J(0)}$ in $(0, x_{I(0)})$, and, for $1 \leq b < n$, choose $1 - y_{J(b)}$ in $(x_{I(b-1)}, x_{I(b)})$. The intervals are disjoint, so the column indices are distinct. These choices give

$$T_n(c) = \begin{pmatrix} c & 1 & \dots & 1 \\ c & c & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \dots & c \end{pmatrix}, \quad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtracting each preceding row from the next, working upwards from the last row, leaves diagonal entries $c, c-1, \dots, c-1$. The same I, J work for every k , whose only effect is multiplication by nonzero row and column factors. The empty minor for $n = 0$ equals 1. Finally, d separated summands would factor every $(d+1) \times (d+1)$ restriction through a d -dimensional space, contradicting the nonzero minor. $\square$

Both clauses are [Lean source](#).

Corollary 1.2 (the same minors modulo every admissible denominator). *For $(p, q, r) = (2, 3, 5)$, the maps I, J in Theorem 1.1 can be chosen so that their minors are invertible over $\mathbb{Z}/B\mathbb{Z}$ for every $B \geq 2$ coprime to 30 and every $k \geq 0$.*

Proof. Every row and column factor is a unit modulo B . The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. The choices of I, J are independent of both B and k . $\square$

The statement is [Lean source](#).

Example 1.3. For $(p, q, r) = (2, 3, 5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

The four entries and the determinant are [Lean source](#).

The indices must be selected. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$K(3, j, 0) = \frac{1}{120} K(0, j, 0) \quad (0 \leq j < 4).$$

These equalities follow by evaluating the height at 3^j and $8 \cdot 3^j$. At $j = 4$ the difference is $-1/19440000$. Thus the leading minors do not supply the arbitrary-order theorem.

Proposition 1.4 (finite sampled cut rank). *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq k \leq m$ write v_k for the length- m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0, m\} \subseteq E}$.*

Proof. List $E = \{k_1 < \dots < k_t\}$. The $t - 1$ differences $v_{k_{j+1}} - v_{k_j} = (1 - c)\mathbf{1}_{\{k_j, \dots, k_{j+1}-1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t . If $k_1 = 0$ and $k_t = m$, then $v_0 = c\mathbf{1}$ lies in the span of the differences (their sum is $(1 - c)\mathbf{1}$), so the rank is $t - 1$. $\square$

The statement is [Lean source](#).

The formula classifies every finite sampled kernel rank after the row and column factors of Theorem 1.1, independently of the third coordinate. An exact logarithm-free algorithm orders the rationals $p^i / r^{\lfloor \log_r p^i \rfloor}$, counts those strictly below each $r^{\lfloor \log_r q^j \rfloor + 1} / q^j$, and applies the displayed correction. Direct elimination on the $\{2, 3, 5\}$ kernel agrees with the formula through order 24; larger leading ranks recorded with the accompanying checker use the formula and exact integer phase comparisons.

The norm matters. For the infinite normalised carry matrix,

$$\inf_{F \text{ of finite separated rank}} \|C - F\|_\infty = (1 - c)/2.$$

Distinct columns are exactly $1 - c$ apart in the supremum norm, by density of the row phases. An approximation below half that distance would give infinitely many bounded, uniformly separated columns in a finite-dimensional space. Compactness excludes this. The constant $(1 + c)/2$ attains the bound.

Finite restrictions have a different behaviour. At $c = 1/5$,

$$T = \begin{pmatrix} 1/5 & 1 \\ 1/5 & 1/5 \end{pmatrix}, \quad A = \begin{pmatrix} 3/10 & 9/10 \\ 1/10 & 3/10 \end{pmatrix}$$

satisfy $\det A = 0$ and $\|T - A\|_{\max} = 1/10 < 2/5$. For the original kernel, truncating the first coordinate at N gives separated rank at most N and

$$\|K - K^{(N)}\|_{\ell^1} \leq \frac{pqr p^{-3N}}{(1 - p^{-3})(1 - q^{-3})(1 - r^{-3})}.$$

This follows from $H(x) > x^3 / (pqr)$. Thus the infinite uniform obstruction coexists with geometric approximation in the summation norm. The scalar irrationality question requires arithmetic information about the actual multiplicities.

2 The two-prime comparison

For distinct primes $p < q$, write $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$. Prime-power divisibility identifies this product with the running LCM. Let $\mathcal{R}_{p,q}$ retain every smooth-number multiplicity, and let $\mathcal{D}_{p,q}$ retain the initial value and one reciprocal at each positive pure-power jump. The following argument is due to Fan [7].

Theorem 2.1 (both two-prime sums). Put $\theta = \log p / \log q$ and $A = \sum_{n \geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Then

$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \quad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \quad (1)$$

Both numbers are transcendental.

Proof. Set $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the p -channel contribute $A = \sum_{n \geq 0} x^n y^{m_n}$. There is one q -power between p^n and p^{n+1} precisely when $\delta_n = 1$, so the other channel contributes

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_{n+1}}, \quad \mathcal{D}_{p,q} = A + B.$$

All these series converge absolutely. Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$, shifting the series for A gives $A - 1 - xA = x(y - 1)B/y$. Therefore

$$B = \frac{p - (p-1)A}{q-1}.$$

At $p^i q^j$ the height is $p^{i+\lfloor j/\theta \rfloor} q^{j+m_i}$, hence

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, each $j \geq 1$ corresponds to $n = \lfloor j/\theta \rfloor$ with $m_n = j - 1$ and $\delta_n = 1$. These identities prove (1).

For the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$, geometric summation gives

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x, y). \quad (2)$$

Bugeaud and Laurent’s Theorem 1.1 applies to these nonzero algebraic x, y : $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$; the zero-shift case goes back to Loxton and van der Poorten [5, 6]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. $\square$

The two-prime reduction expresses both sums through one value. The third prime leaves the binary carry of Section 1. Its rank theorem concerns that function; the scalar arithmetic depends on the multiplicities introduced next. The de-duplicated irrationality assertion is historical [3].

3 Exact multiplicities and normalised tails

Proposition 3.1 (the running least common multiple). Let p, q, r be pairwise distinct primes and $x \geq 1$. Then $L(x) = H(x)$.

Proof. Every smooth $n \leq x$ has prime exponents bounded by the corresponding integer logarithms, so $n \mid H(x)$. Conversely the three maximal pure powers occur among those smooth numbers; their product divides the running LCM because they are pairwise coprime. $\square$

The statement is [Lean source](#). The same definition gives

$$x^3/(pqr) < H(x) \leq x^3. \quad (3)$$

Both inequalities are [Lean source](#). The height is constant between consecutive pure-power boundaries and gains a factor equal to the prime at each boundary. If $F(H)$ is a height fibre in a finite exponent box $\mathcal{B}$, exact regrouping gives

$$\sum_{(i,j,k) \in \mathcal{B}} K(i,j,k) = \sum_H \frac{\#F(H)}{H}. \quad (4)$$

Each term in $F(H)$ equals $1/H$, which proves the identity. The multiplicities remain present in every subsequent infinite sum. The regrouping is [Lean source](#).

From now on let $P = \{2, 3, 5\}$ and $S = \mathcal{R}_P$. For $a \geq 0$ define

$$s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{1}{H(x)}, \quad T_a = \sum_{j \geq 0} s_{a+j}, \quad h_a = \frac{H(2^a)}{2}, \quad X_a = h_a T_a.$$

Thus T_a is the raw tail and X_a its normalised state. The literal radix and forcing digit are

$$b_a = \frac{H(2^{a+1})}{H(2^a)}, \quad m_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{H(2^{a+1})}{2H(x)}. \quad (5)$$

Lemma 3.2 (integer forcing and the four radices). *For every $a \geq 0$, m_a is a positive integer and $b_a \in \{2, 6, 10, 30\}$.*

Proof. If $x < 2^{a+1}$, the two-exponent of $H(x)$ is at most a , while the other exponents are bounded by those of $H(2^{a+1})$. Hence $2H(x) \mid H(2^{a+1})$. Each forcing summand is an integer, and the shell contains 2^a . Between consecutive two-powers there is at most one three-power and at most one five-power. Their optional factors, together with the terminal factor two, give the four radices. $\square$

Both clauses are [Lean source](#).

Proposition 3.3 (the actual recurrence and a polynomial bound). *The series defining S, T_a converge. For every $a \geq 0$,*

$$X_{a+1} = b_a X_a - m_a, \quad 0 < X_a \leq \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \geq 1$, either some BX_a is integral and all later states are integral, or $\text{dist}(BX_a, \mathbb{Z}) \geq 1/31$ at arbitrarily large indices.

Proof. A dyadic shell contains at most one two-exponent for each pair of three- and five-exponents. Each of the latter lies between 0 and a , so its count is at most $(a+1)^2$. By (3), $s_a \leq 30(a+1)^2/8^a$. Therefore

$$X_a \leq 15(a+1)^2 \sum_{j \geq 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This proves convergence and the bound. Splitting the first shell gives the recurrence because $m_a = h_{a+1}s_a$ and $h_{a+1} = b_a h_a$. All four clauses are [Lean source](#).

Put $Y_a = BX_a$. If all sufficiently late distances are strictly below $1/31$, write $Y_a = z_a + e_a$ with $z_a \in \mathbb{Z}$ and $|e_a| < 1/31$. Then $e_{a+1} - b_a e_a$ is an integer of absolute value less than one, so $e_{a+1} = b_a e_a$. The factors $b_a \geq 2$ force a bounded such error to be zero. An integral state propagates by the integer recurrence. $\square$

The constant above is a simple bound for the cap used below. The stronger source estimate $\tilde{Q}(n) = (1210n^2 + 9130n + 18847)/11979$ uses the exclusion of three successive two-jumps and has a separate ordinary proof. The present cap suffices for the criterion; the sharper argument belongs in the extended record rather than in this proof.

Theorem 3.4 (actual rationality-to-reduced-carry bridge). *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \geq a_0 = u + 1 + 2v + 3w$,*

$$d_a = BX_a \in \mathbb{Z}_{>0}, \quad d_{a+1} = b_a d_a - Bm_a, \quad d_a \leq 90B(a+1)^2.$$

Proof. For $a \geq 1$, strict boundary clearing gives

$$X_a = h_a S - Z_a, \quad Z_a = \sum_{\substack{x \text{ smooth} \\ x < 2^a}} \frac{h_a}{H(x)} \in \mathbb{Z}. \quad (6)$$

The two-exponent of h_a is $a - 1$. Moreover $a \geq 2v$ gives $2^a \geq 3^v$, and $a \geq 3w$ gives $2^a \geq 5^w$. Thus $2^u 3^v 5^w \mid h_a$ after the stated onset, and $BX_a = h_a A / (2^u 3^v 5^w) - BZ_a$ is integral. Positivity, the recurrence and the bound follow from Proposition 3.3. $\square$

The statement is [Lean source](#).

4 A window test and the remaining arithmetic

The actual digits in (5) define

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

Induction on h gives

$$X_{\ell+h} = W_{\ell,h} X_\ell - F_{\ell,h}. \quad (7)$$

Let $\text{lpr}_W(t)$ be the representative in $\{1, \dots, W\}$ of $t \bmod W$, representing a zero residue by W . Throughout this section use the valid cap

$$K(B, a) = 90B(a+1)^2. \quad (8)$$

Lemma 4.1 (finite endpoint obstruction). *If d is a positive integer with $d \leq K$ and $d \equiv -BF \pmod{W}$, where $W \geq 1$, then $\text{lpr}_W(-BF) \leq K$.*

Proof. Every positive representative of the residue is at least its least positive representative, so $\text{lpr}_W(-BF) \leq d \leq K$. $\square$

The statement is [Lean source](#).

Theorem 4.2 (the actual window criterion). *The number S is irrational if and only if*

$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0, h \geq 1 \text{ such that } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell + h). \end{aligned} \quad (9)$$

Proof. If S is rational, Theorem 3.4 supplies a positive integral carry $d_a = BX_a$ after its onset. Choose a window in (9) beyond that onset. Equation (7) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contradicting Lemma 4.1 and the valid cap.

Conversely, let S be irrational and fix $B, \ell \geq 1$. Equation (6) makes BX_ℓ nonintegral. Put $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0, 1)$. Since $W_{\ell,h} \geq 2^h$ and $X_{\ell+h} = O((\ell + h + 1)^2)$, for all sufficiently large h the number $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0, 1)$. The window identity then gives the exact equality

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its exponentially growing first term eventually exceeds $K(B, \ell + h)$. This proves (9), in fact from each fixed start. $\square$

The equivalence, at the literal smooth-number series, is [Lean source](#).

The validity of the cap is needed in the rational direction; an upper polynomial growth condition by itself would not suffice. The zero cap, for example, makes positive-residue escape automatic. The criterion identifies the target in window coordinates. The unresolved point is the source-specific exclusion below.

Problem 4.3 (nonintegrality of the actual reduced tails). Prove that for every $a \geq 1$ and every integer $B \geq 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \quad (10)$$

By (6), one integral value would make S rational; the bridge proves the reverse implication after clearing its denominator. Thus the pointwise formulation is equivalent to the irrationality question.

The information the remaining argument must use. The bounded-radix alternative permits the integral branch. Infinite rank and the modular minors of Corollary 1.2 constrain the kernel, while a scalar argument must also use its exact multiplicities. In the existing three-channel rigidity result, preservation of every complete same-channel block makes a perturbation a difference of three channel potentials. Two zero anchor transitions make that potential constant. A bounded rationalising lift does not automatically supply those unweighted block identities: its actual block defect is weighted. A proof using this route must establish that extra source-faithful condition. The extended record retains the exact countermodels and the other proposed representations at their stated scopes.

Statements and declarations

Proof sources. The two-prime argument uses the cited Hecke–Mahler value theorem. The arbitrary-order kernel theorem, the actual recurrence and the rationality bridge have the formal source locations recorded below. The finite cut-rank argument and the modular corollary are proved above; no additional formal verification claim is made for the latter.

Artefact and data availability. The [pinned source revision](#) is f36a98bf3d3e. The extended reasoning record and the accompanying source retain the detailed shell bounds, finite experiments and alternative representations with their exact hypotheses. Supplied later-source labels and public links remain separate source coordinates.

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A Guide to the principal formal statements

The arbitrary-order minor and non-separability statements are `exists_uniform_nonsingular_threePrimeKernel_minor_of_prime` and `not_finite_separable_threePrimeKernel` in `ErdosProblems/Erdos269/KernelCarryRank.lean`. The same file contains the leading-four identities. The interval-difference algebra behind the finite cut-rank formula is in `ErdosProblems/Erdos269/FiniteCutRank.lean`; that file by itself is not a formal proof of the full arithmetic normal form.

The running-LCM and finite shell statements are in `ErdosProblems/Erdos269/ThreePrimeRunningLcm.lean`. The actual bridge is `exists_reducedCarry_of_value_eq_rat` in `ErdosProblems/Erdos269/RationalityCarryBridge.lean`. The finite residue argument is in `ErdosProblems/Erdos269/ResidueEscape.lean`, and its window composition is in `ErdosProblems/Erdos269/RestrictedFloorSum.lean`. The exact block-potential classification and its anchor condition are in `ErdosProblems/Erdos269/ThreeChannelBlockRigidity.lean`. These source statements retain their displayed hypotheses.

References

- [1] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 65. For a possibly infinite prime set Q , the page states the infinite- Q irrationality and asks what happens for finite Q with more than one element.
- [2] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:[10.1017/CBO9780511897184.009](https://doi.org/10.1017/CBO9780511897184.009).
- [3] P. Erdős, *Letter to the Editor* (written 1 January 1973), *Fibonacci Quart.* **12** (1974), no. 4, p. 335. The letter poses the full series as a conjecture and asserts irrationality after retaining only the distinct running-LCM values.
- [4] T. F. Bloom, *Erdős Problem #269*, erdosproblems.com/269, accessed 28 July 2026 (page displays “last edited 28 December 2025”). The current record labels the finite-support problem open, cites [ErGr80, p. 65] and [Er88c, p. 106], routes to the 1974 letter on p. 335, records the infinite-prime and de-duplicated variants, and explicitly describes its status as the website owner’s present assessment rather than a literature-completeness guarantee.

- [5] Y. Bugeaud and M. Laurent, *Transcendence and continued fraction expansion of values of Hecke–Mahler series*, Acta Arith. **209** (2023), 59–90, doi:10.4064/aa220323-18-1; [authors' publisher-layout PDF](#), [arXiv:2203.12901v1](#). Theorem 1.1 is on journal p. 61 (publisher-layout PDF p. 3); the authors note there that its $\rho = 0$ case was already obtained by Loxton and van der Poorten.
- [6] J. H. Loxton and A. J. van der Poorten, *Arithmetic properties of certain functions in several variables III*, Bull. Austral. Math. Soc. **16** (1977), 15–47.
- [7] S. Fan, comment on Erdős Problem #269, [erdosproblems.com forum, thread 269](#), 26 June 2026. The comment gives the two-channel factorisation, the Hecke–Mahler reduction, and the transcendence conclusion for $|P| = 2$; follow-up comments there note the extension to arbitrary coprime pairs.